Institut de Recherche pour le Développement
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LOCMARCA 2026 · COURSE 2

Numerical models

Ocean, atmosphere, climate: what a model computes, and what it can tell you

LOCMARCA 2026

GOALS

What you should take away

1

Understand

What a model computes, and the four ideas behind any model: resolution, time step, processes, grid.

2

Use

Pick the right kind of model output for a question: process study, reanalysis, forecast or projection.

3

Question

Know the limits of a model, and always check it against observations before you trust it.

Not a modelling course: a way of reading **model outputs** like the ones you will open in TP3a and TP3b.

ROADMAP

Five parts

1

What is a model?

Equations on a grid, a run step by step, a long history

2

Key ideas

Scales, resolution, time step, processes, grids

3

What for?

Four uses, process studies, choosing a model

4

Global vs regional

Nesting, scenarios, and the limits of models

5

Climate models

Coupled Earth system models, CMIP6, ESGF

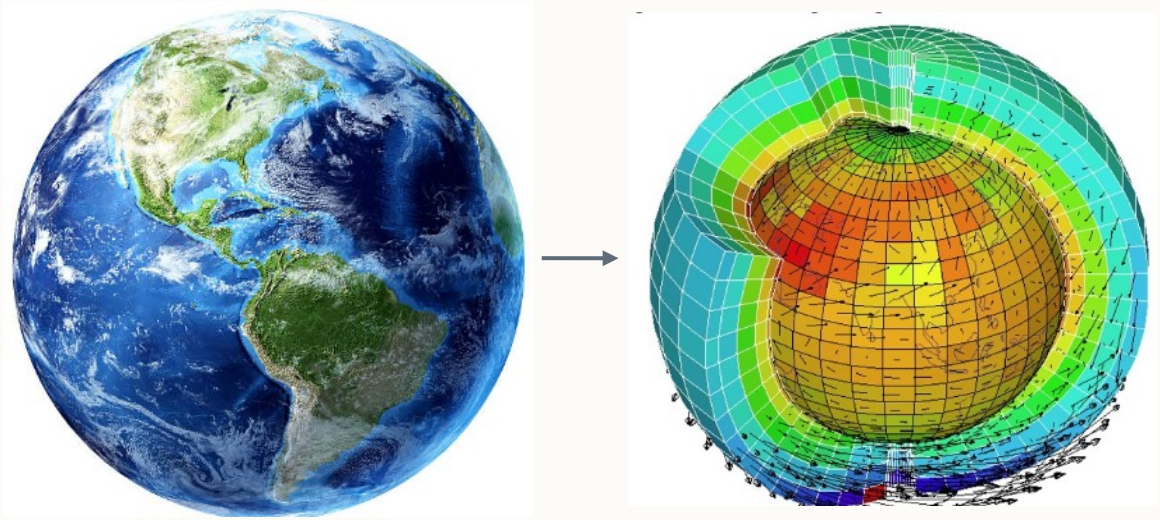
Today: Course 2. **Tomorrow:** TP3a (NEMO, global) and TP3b (CROCO, regional), then the TP4 mini-project.

PART 1 · WHAT IS A NUMERICAL MODEL?

A model solves the laws of physics on a grid, one time step after another.

It does not measure anything: it computes what the equations predict, from what we give it.

What is ocean modelling?



The real ocean

Its model: the ocean cut into grid cells

The ocean counterpart

of the better-known weather (atmospheric) models.

A computer code

that computes the state of the ocean at a later time, from its state at an initial time and all the forces acting on it.

Numerical approximations

of the equations of physics: for most ocean general circulation models (OGCMs), the primitive equations.

The primitive equations, in plain words

advection Coriolis pressure gradient forcing terms diffusive terms

$$\frac{\partial u}{\partial t} + \vec{\nabla}.(\vec{\mathbf{v}}u) - fv = -\frac{\partial \phi}{\partial x} + \mathcal{F}_u + \mathcal{D}_u$$

$$\frac{\partial v}{\partial t} + \vec{\nabla}.(\vec{\mathbf{v}}v) + fu = -\frac{\partial \phi}{\partial y} + \mathcal{F}_v + \mathcal{D}_v$$

$$\frac{\partial C}{\partial t} + \vec{\nabla}.(\vec{\mathbf{v}}C) = \mathcal{F}_C + \mathcal{D}_C$$

$$\frac{\partial \phi}{\partial z} = -\frac{\rho g}{\rho_0}$$

$$\vec{\nabla}.\vec{\mathbf{v}} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

$$\rho = \rho(T, S, P)$$

$$\frac{\mathrm{d}\zeta(\mathbf{x_H}, t)}{\mathrm{dt}} = w(z = \zeta) \qquad p(\mathbf{x_H}, z = \zeta, t) = p_{atm} \qquad \mathbf{v}(z = -H) = \mathbf{0}$$

Momentum: currents change by advection, Coriolis, pressure gradients, forcing and friction

Tracers (T, S): carried by the currents and mixed

Hydrostatic balance: pressure = weight of the water above

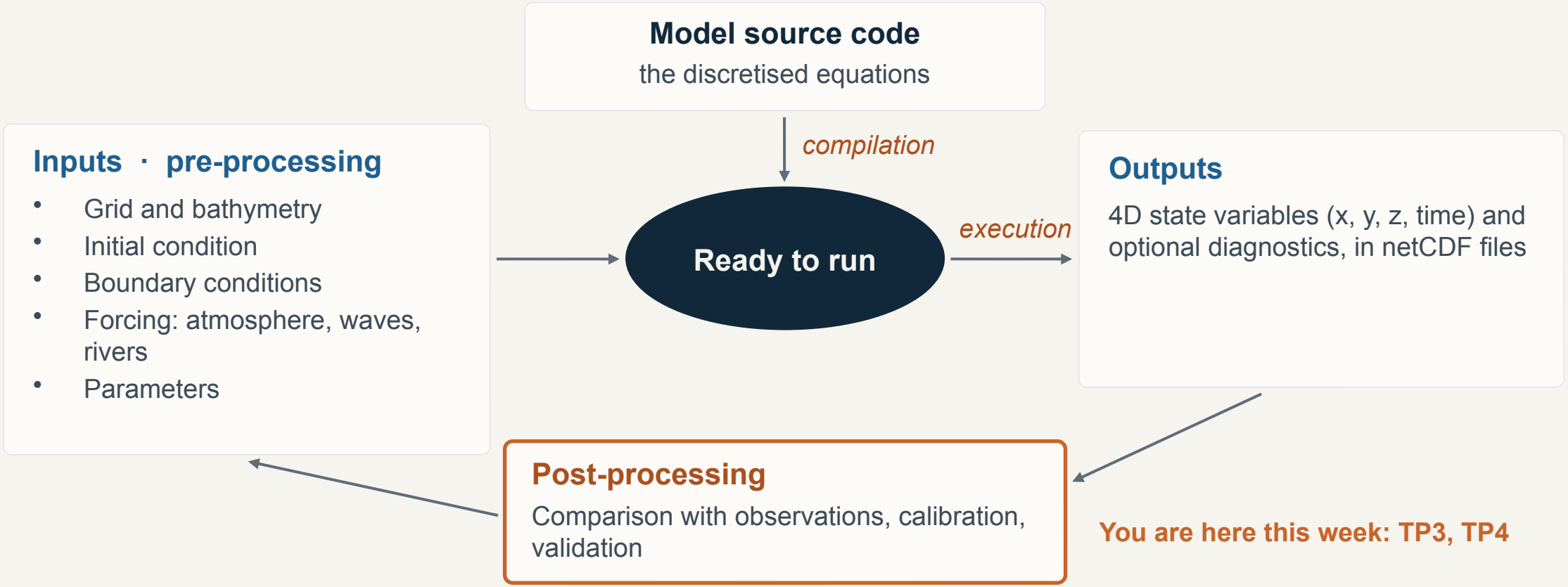
Continuity: water is neither created nor destroyed

Equation of state: density from temperature, salinity, pressure

Boundary conditions: at the surface and at the sea floor

F, D: forcing and the parameterised effect of the small scales (Part 2).

How a model run works



One method, three families of models

Atmosphere

TIME SCALE

Changes in hours; memory of about two weeks

USED FOR

Weather forecasts, atmospheric reanalyses (ERA5, the winds of TP2a)

WRF, ECMWF IFS, ARPEGE

Ocean

TIME SCALE

Changes in days to centuries; stores heat and carbon

USED FOR

Currents, upwelling, El Niño, the Costa Rica Dome, ocean reanalyses (GLORYS12)

NEMO (TP3a), CROCO (TP3b), MOM, HYCOM

Climate

TIME SCALE

Ocean + atmosphere + sea ice + land, coupled

USED FOR

Past and future climate over decades to centuries, scenarios

IPSL-CM6A-LR, CNRM-CM6, CESM2

A long history

1963

Kirk Bryan, GFDL

First numerical ocean circulation models

1967

Bryan & Cox; Holland

The first 3D ocean model (GFDL); Scripps

1980s

MOM, POM, OPA, HYCOM…

Many ocean models are born

1994–2000

SCRUM, then ROMS

Terrain-following (sigma) models

2016

CROCO

Coastal and Regional Ocean COmmunity model

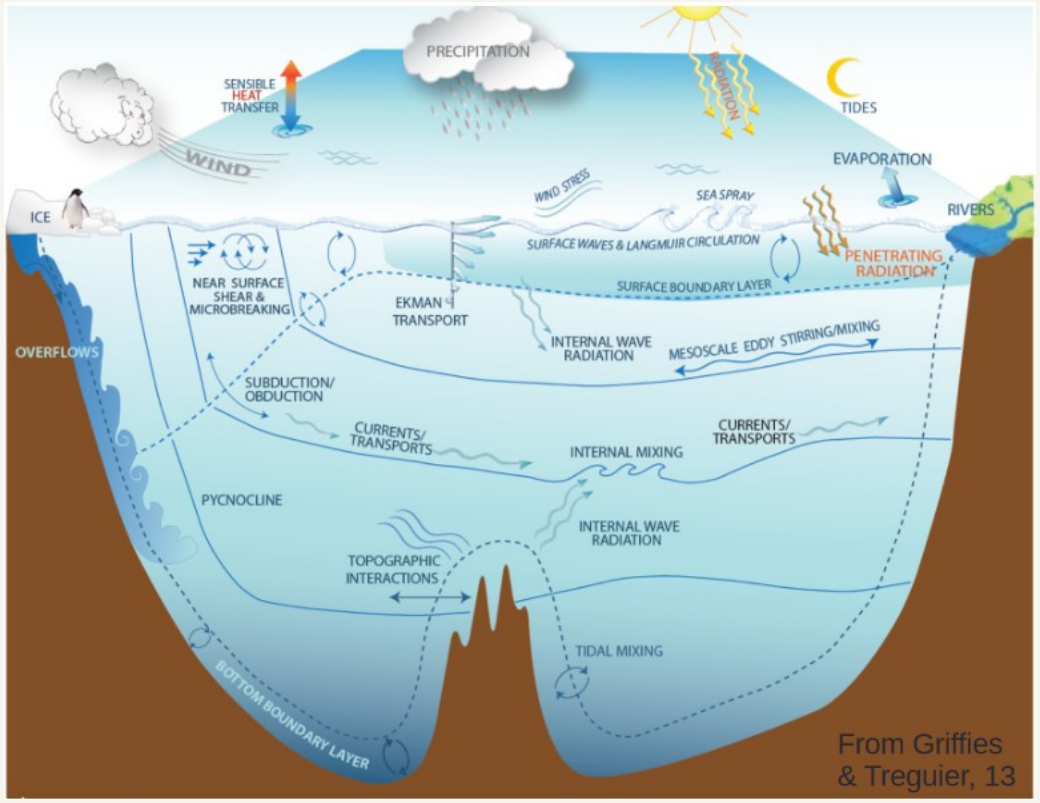
NEMO · TP3a

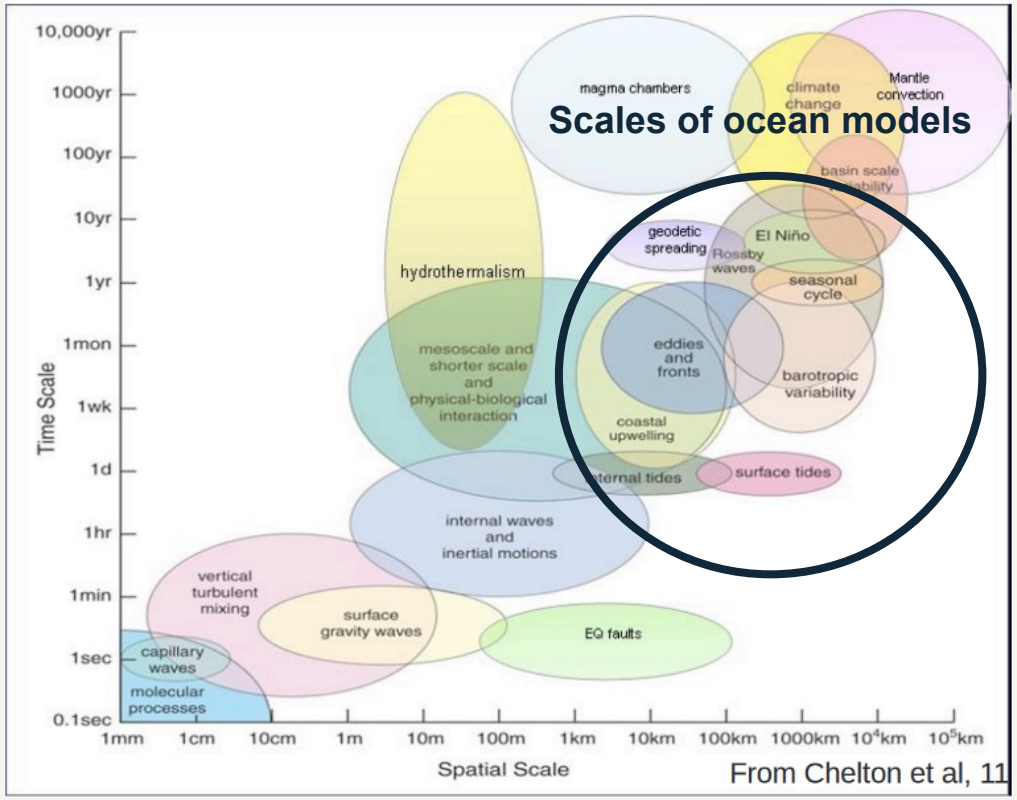
Born from the French model OPA; z-levels; global and basin scales, also inside climate models.

CROCO · TP3b

The last of the sigma family: POM → SCRUM → ROMS → CROCO; regional and coastal scales.

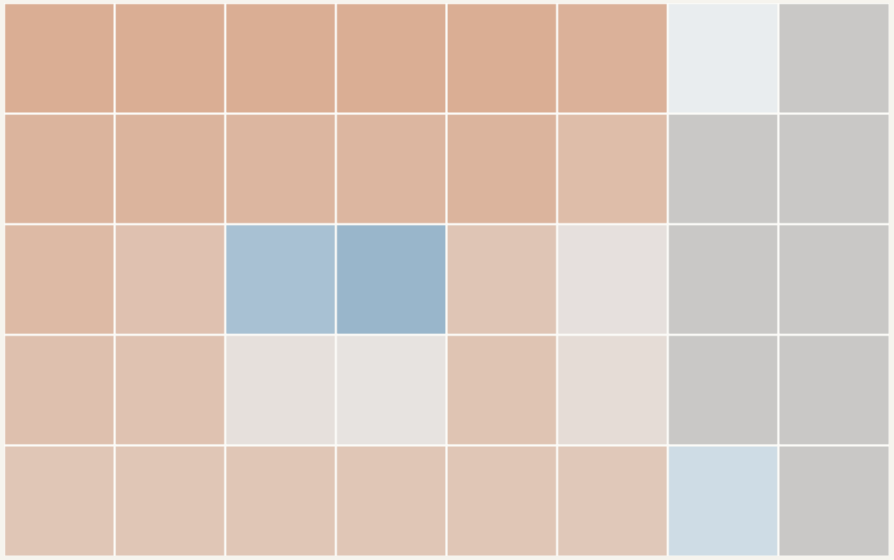
A zoo of processes and scales



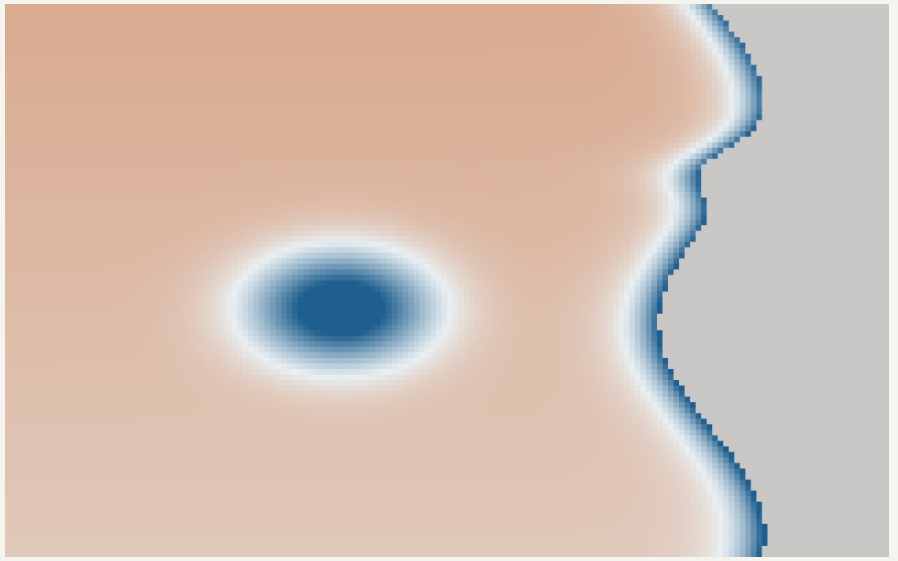


Left: Griffies & Treguier (2013). Right: Chelton et al. (2011).

Resolution: one value per grid cell



1° grid (≈ 110 km) · NEMO, TP3a



≈ 5 km grid · CROCO, TP3b

5–10

grid cells are needed to represent a feature

Idealised field: a cold eddy offshore and a cold strip along the coast (land in grey).

From an equation to a computation

1 **A simple equation**

Temperature T carried by a current u0 along x

$$\frac{\partial T}{\partial t} + u_0 \frac{\partial T}{\partial x} = 0$$

2 **Derivatives become differences**

between neighbouring grid points (δx) and time steps (δt)

$$\frac{\partial T}{\partial x} \sim \frac{T(x+\delta x) - T(x)}{\delta x} = \frac{T_{i+1,n} - T_{i,n}}{\delta x}$$

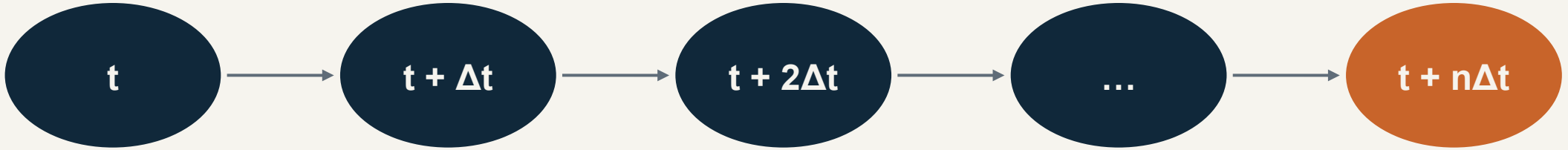
$$\frac{\partial T}{\partial t} \sim \frac{T(t+\delta t) - T(t)}{\delta t} = \frac{T_{i,n+1} - T_{i,n}}{\delta t}$$

3 **A rule the computer repeats**

the new T at point i, from the old T at i and i + 1

$$T_{i,n+1} = T_{i,n} - \delta t \left(u_0 \frac{T_{i+1,n} - T_{i,n}}{\delta x} \right)$$

Time integration: step by step



state → compute the tendencies from the equations → new state

≈ 1 h

time step of a 1° global ocean model

minutes

time step of a ≈ 5 km regional model

× 8

cost when the grid spacing is divided by 2

Processes: resolved or parameterised

Resolved

Larger than a few grid cells: computed by the equations

- Large-scale currents and gyres
- Upwelling driven by the wind
- El Niño, Kelvin and Rossby waves
- Eddies and fronts, but only if the grid is fine enough (≈ 5 km yes, 1° no)

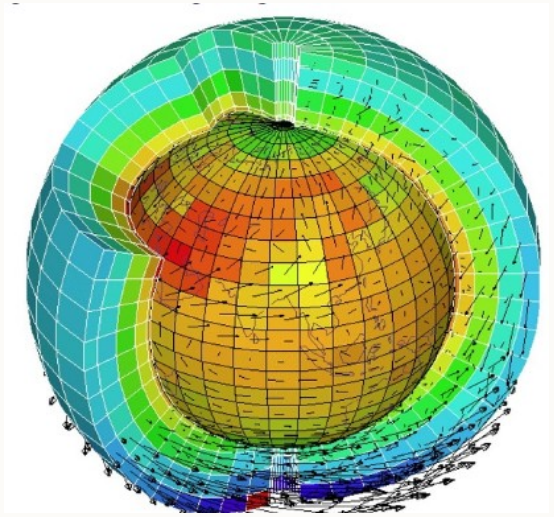
Parameterised

Smaller or faster than the grid: replaced by a formula

- Vertical mixing and turbulence (depth of the mixed layer)
- Convection
- Eddies in a 1° model
- In the atmosphere: clouds, rain, radiation

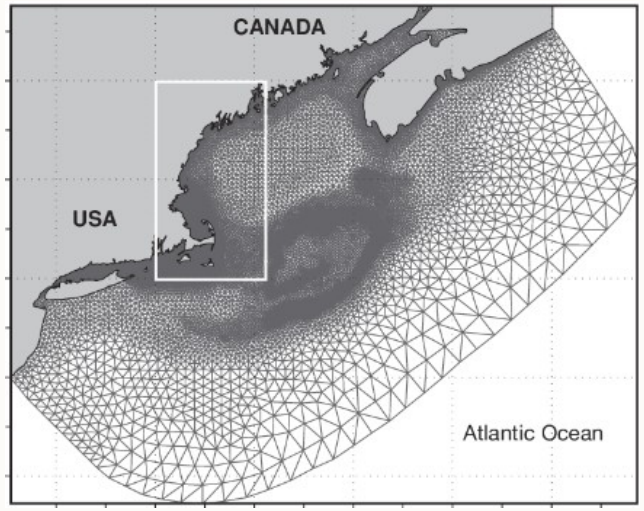
Parameterisations are the main reason why two models of the same ocean give different answers.

Grid types: a diversity of horizontal grids



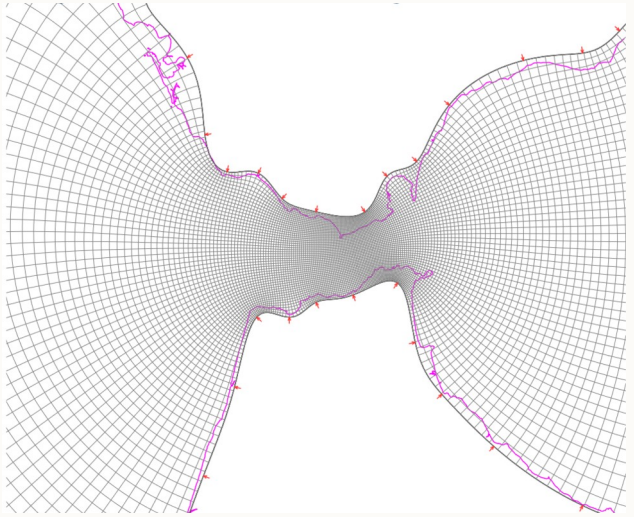
Logically rectangular

curvilinear; finite differences or volumes; open ocean, large scales (NEMO, CROCO)



Unstructured: triangles

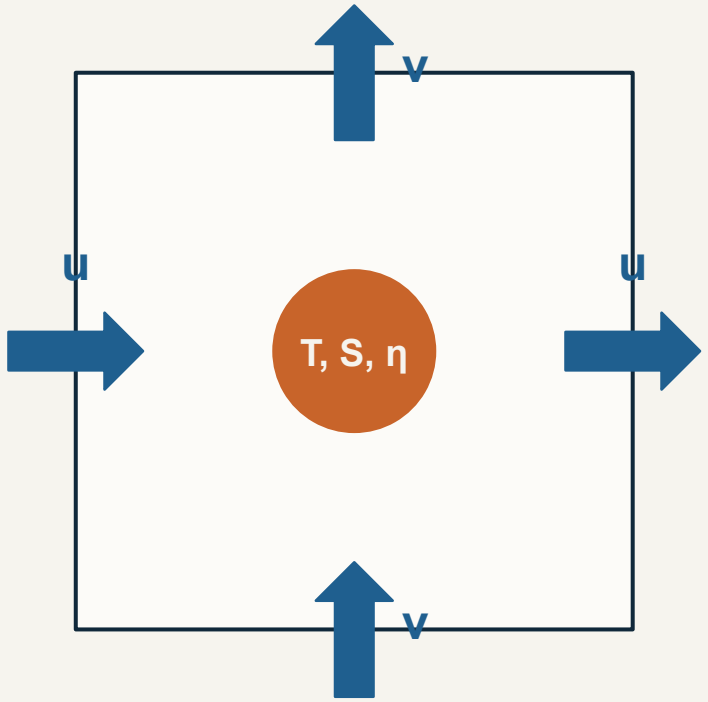
finite volumes or elements; coasts and estuaries



Curvilinear, fitted to a coast

a grid following a strait (Izogrid, Shchepetkin)

Grid types: where the variables live



One cell of an Arakawa C-grid

What it means for your files

u, v and T do not share the same coordinates

CROCO: temp on (eta_rho, xi_rho), u on (eta_rho, xi_u), v on (eta_v, xi_rho)

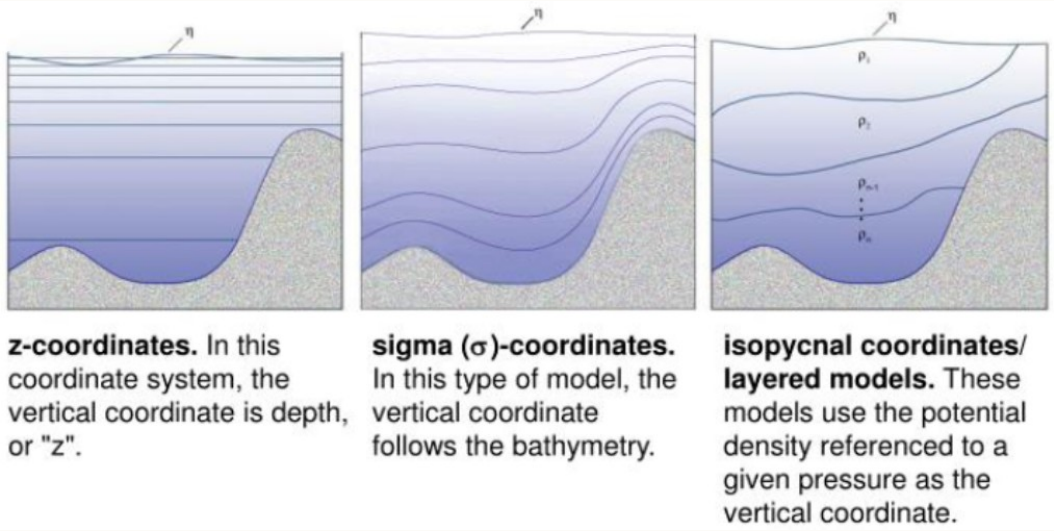
Interpolate before you combine

Bring u and v to the T points before plotting arrows on a temperature map (TP3b)

The grid may be curvilinear

2D longitude and latitude arrays instead of 1D axes, as in CROCO and near the poles in NEMO

Grid types: the vertical levels



z-coordinates. In this coordinate system, the vertical coordinate is depth, or "z".

sigma (σ)-coordinates. In this type of model, the vertical coordinate follows the bathymetry.

isopycnal coordinates/ layered models. These models use the potential density referenced to a given pressure as the vertical coordinate.

z-levels · NEMO, TP3a

Fixed depths: easy to read; the sea floor is a staircase.

Sigma levels · CROCO, TP3b

They follow the sea floor. A level is not a depth: interpolate.

Isopycnal · e.g. HYCOM

They follow density: good for the ocean interior.

Four uses, four kinds of output



Process studies

Understand how the ocean works, with experiments nature does not allow

e.g. the regional runs of TP3b



Reanalyses

Model + observations (data assimilation): the best description of the past

e.g. GLORYS12 (TP2), ERA5



Forecasts

Start from today's state and run a few days ahead

e.g. weather, Copernicus Marine forecasts



Projections

Decades to centuries ahead, under a scenario of future emissions

e.g. CMIP6 (Part 5)

Advantages: cheap compared with ships, a complete 4D view (x, y, z, time), hypotheses can be tested. **Limits:** approximate equations and numerics, parameterisations (Part 4).

Process studies: virtual experiments

Control run

The model with all its forcings, first checked against observations

VS

Sensitivity run

The same run with one thing changed: e.g. no wind jets through the Central American gaps

=

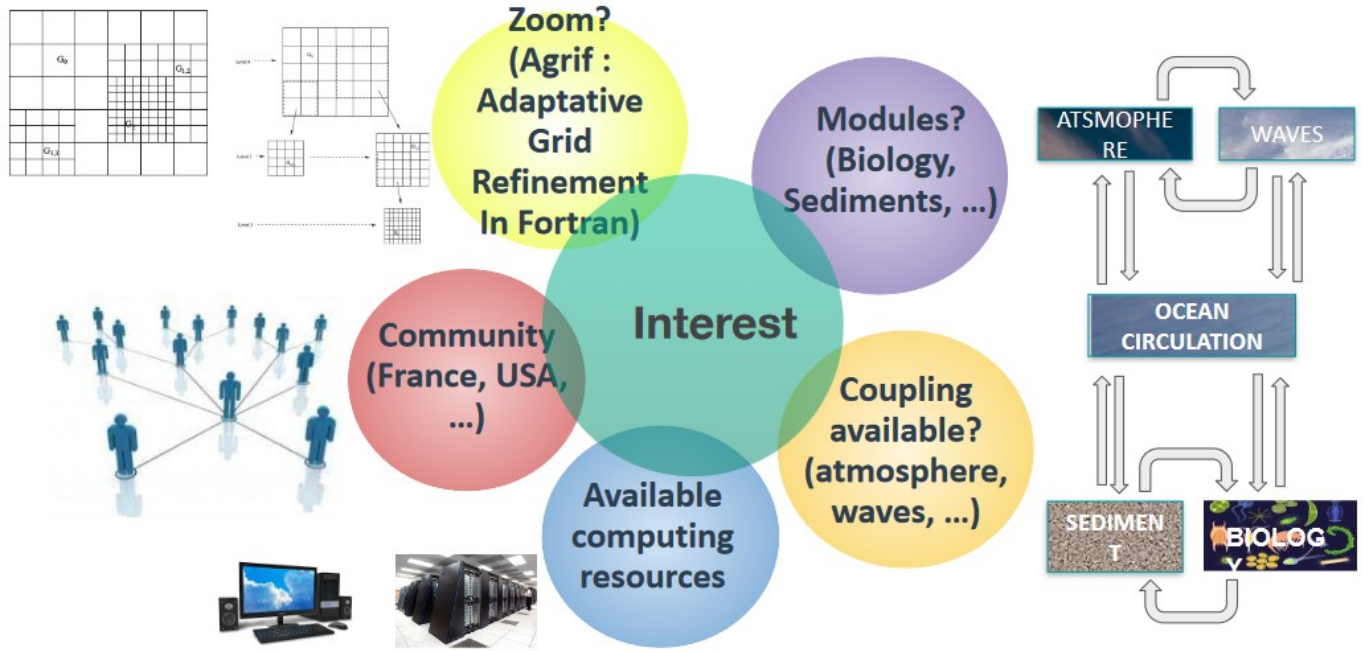
Difference

Experiment − control = the effect of that one process

Other experiments

- Switch the tides or a river on and off
- Change the resolution: what does a 5 km grid add to a 1° grid? (TP3b vs TP3a)
- Change a parameterisation and compare with observations

In practice: which ocean model?

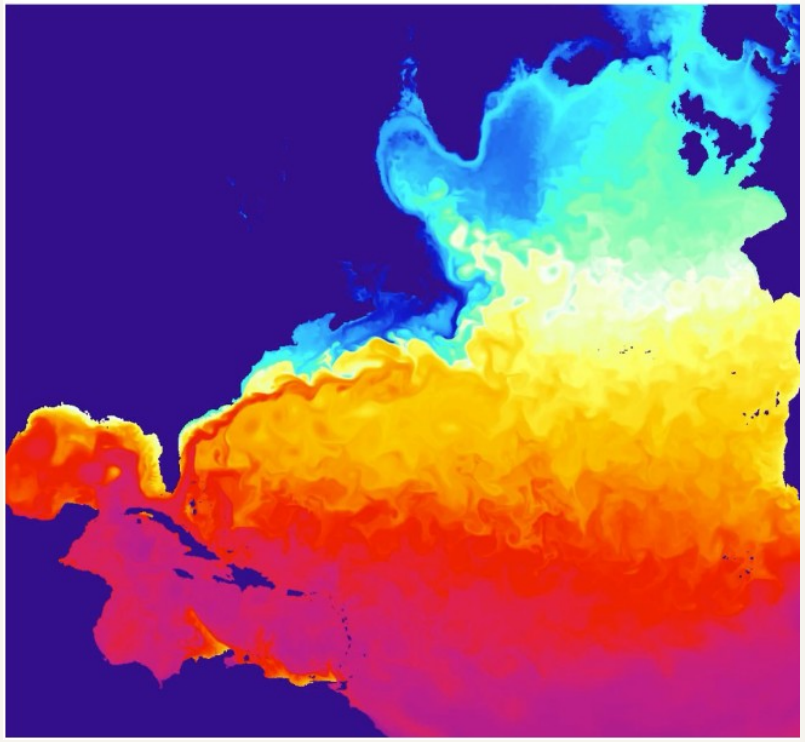


Questions to ask

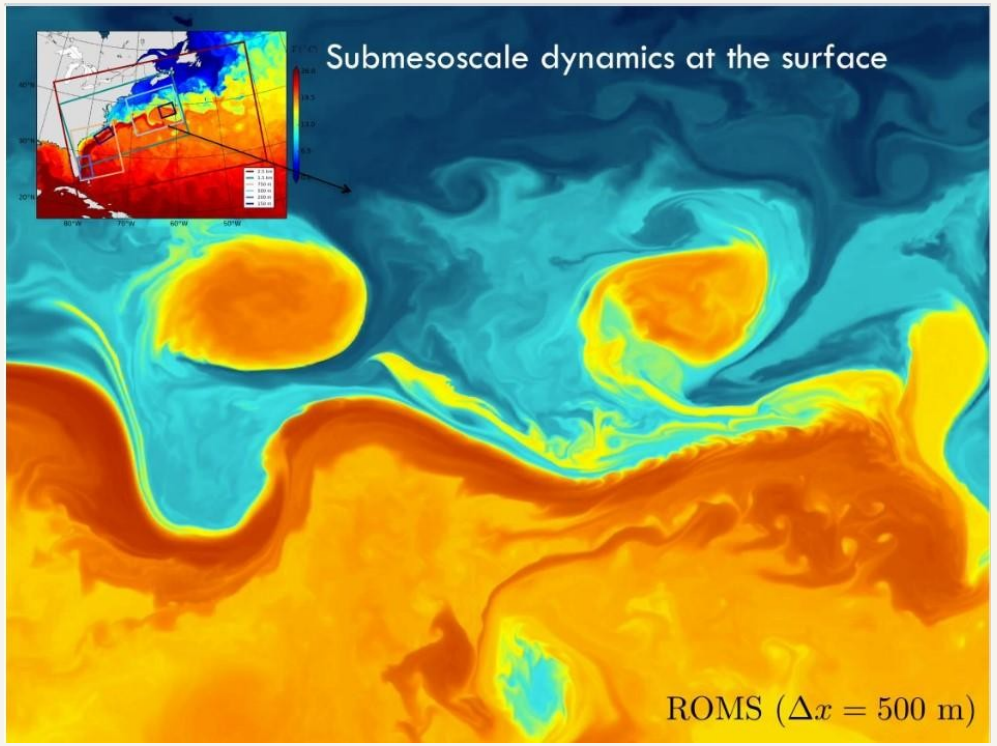
- Which scales and region?
- A community around you?
- Computing resources: laptop or HPC (Kabre)?
- Modules: biology, sediments?
- Coupling: atmosphere, waves?
- Zooms (AGRIF in CROCO)?

As a user: the model is often already chosen. Read its documentation.

A taste of what ocean models can do



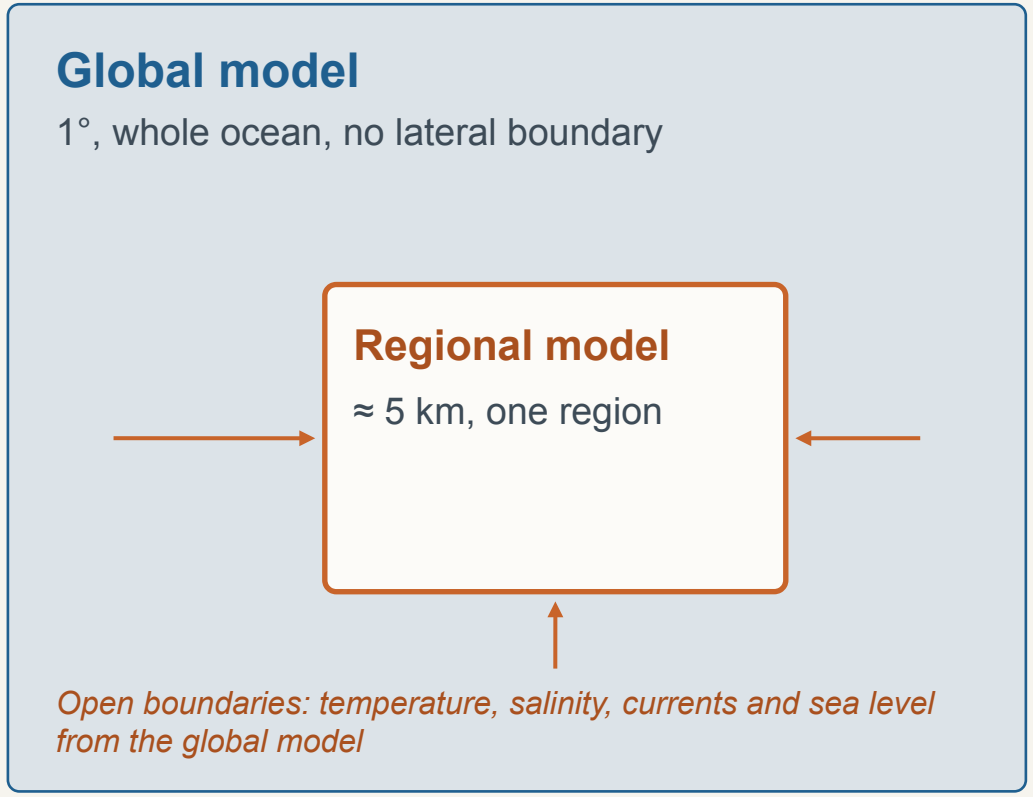
Coupled North Atlantic: ocean 6 km, atmosphere 18 km



Submesoscale eddies: ROMS, Δx = 500 m

Courtesy J. Gula, LOPS, Brest

Global vs regional: nesting



Model	Global	Regional
Example this week	NEMO, 1° (TP3a)	CROCO, ≈ 5 km (TP3b)
Domain	Whole ocean	One region
Boundaries	None	From a larger model
Good for	Basins, El Niño, climate	Coasts, upwelling, eddies
Cost per km²	Low	High

Scenarios: a projection is not a forecast

SSP1-2.6	SSP2-4.5	SSP3-7.0	SSP5-8.5
Low emissions	Intermediate	High	Very high
+1.8 °C	+2.7 °C	+3.6 °C	+4.4 °C

Global surface warming, 2081–2100 vs 1850–1900, best estimate (IPCC AR6, WG1).

Projection: “if the world follows this path, the climate of 2070 will look like this, on average.” **Not** “it will rain in Heredia on 3 March 2070.”

PART 4 · LIMITS

Six limits to keep in mind

1 Resolution

Nothing smaller than a few grid cells: coasts, bays, narrow upwelling

2 Parameterisations

Simplified, tuned formulas for what the grid cannot resolve

3 Forcing and boundaries

Errors in the wind or at the open boundaries go into the ocean

4 Initial state and drift

A model needs time to adjust (spin-up) and can slowly drift

5 Systematic biases

e.g. many climate models are too warm in tropical upwelling regions

6 A free run is not the real weather

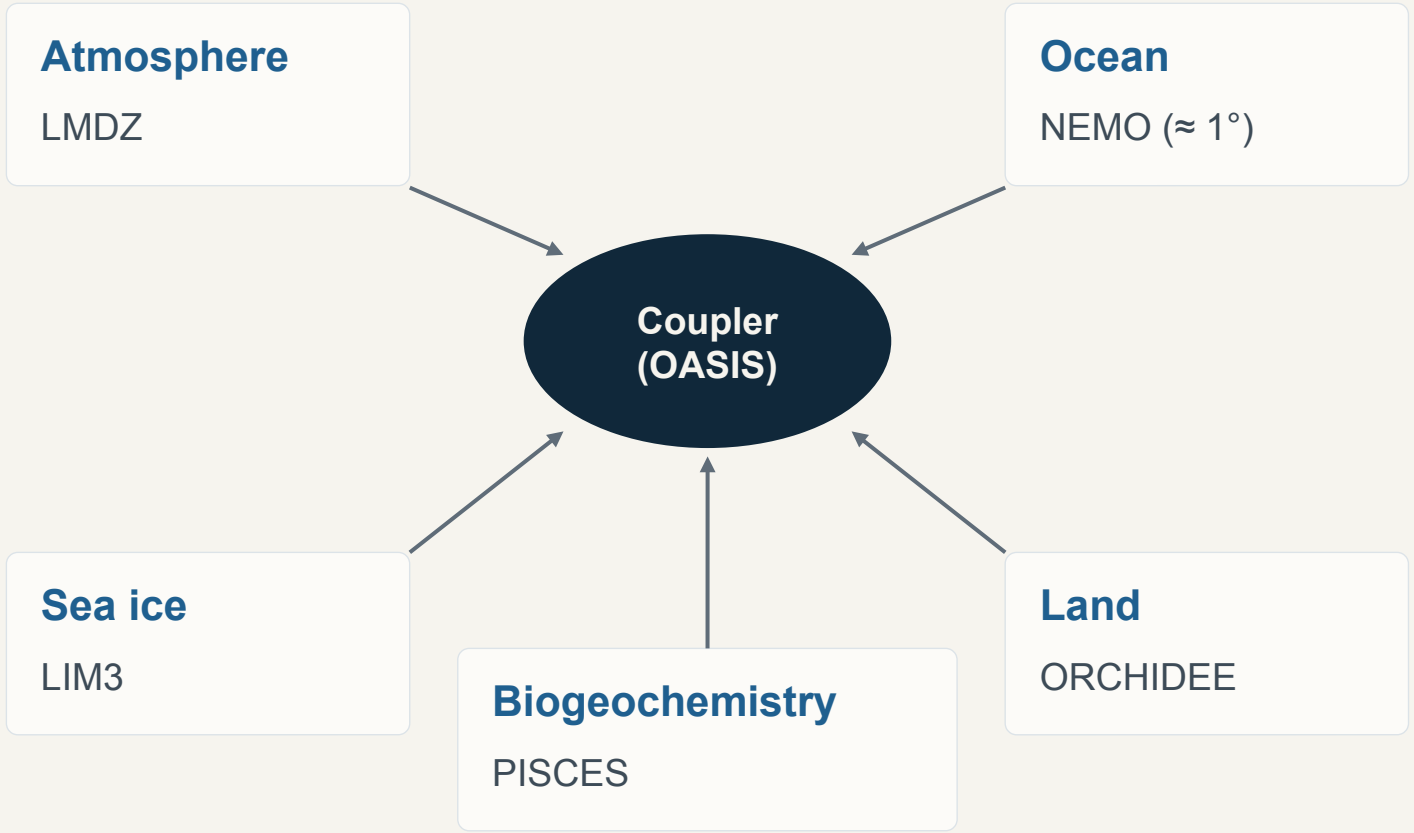
Its eddies are not on the real day: compare statistics, not snapshots

PART 4 · LIMITS

A model output is not an observation. Compare it with data before you trust it.

“All models are wrong, but some are useful.” (G. Box) · TP3b and TP4: models vs observations

Climate models couple the Earth system



IPSL-CM6A-LR, one of the CMIP6 climate models

Exchanged through the coupler

- Heat
- Fresh water: rain, evaporation, rivers, ice melt
- Momentum: the wind stress
- Carbon (CO_2)

Link with this week: NEMO (TP3a) and PISCES (TP2c) are the same models, inside a climate model.

Climate model outputs: CMIP6 on ESGF

tos	Omon	IPSL-CM6A-LR	ssp245	r1i1p1f1	gn	201501-210012	.nc
variable	**table**	**model**	**experiment**	**member**	**grid**	**period**	
sea surface temperature	ocean, monthly	source_id	scenario SSP2-4.5	one run of the ensemble	native model grid	Jan 2015 – Dec 2100	

CMIP6

About 50 modelling groups, one common protocol: same experiments, same variable names. The basis of the IPCC reports, free to download.

ESGF

The federation of data nodes that distributes CMIP6, e.g. the IPSL node:

esgf-node.ipsl.upmc.fr/projects/esgf-ipsl/

One run is not enough: ensembles

Internal variability

The climate is chaotic: the same model, started from slightly different states, gives different El Niño years.

Several members: r1, r2, r3…

Model uncertainty

Different models make different choices (resolution, parameterisations) and give different answers.

Several models: the CMIP6 spread

Your practice

Look at the ensemble mean and its spread, never at one run. Check the historical runs against observations first.

Mean ± spread, validated

TAKE-AWAY

Five questions to ask any model output

1. **Which model, at which resolution?** Grid spacing, number of levels: what is too small to be seen?
2. **Which kind of run?** Free run, reanalysis, forecast or projection under which scenario?
3. **Resolved or parameterised?** Is the process I study computed, or replaced by a formula?
4. **Which grid?** Where are T, u and v? z-levels or sigma levels? 1D or 2D coordinates?
5. **Compared with observations?** What are its known biases in my region? One run or an ensemble?

NEXT

This week, you will use all of this

TP3a · today

NEMO, global, 1°: z-levels, longitudes 0–360, a section across the Costa Rica Dome

TP3b · tomorrow

CROCO, regional, ≈ 5 km: sigma levels, the C-grid, CROCO vs NEMO vs OSTIA

TP4 · mini-project

Your own model–data comparison, on a region of your choice

Questions?